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Harnessing Offshore Renewable Energy from High-Intensity Focused Ocean Waves Using 3D-Printed Undulating Porous Structures

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Abstract

This paper presents a modular seabed infrastructure system that concentrates ocean wave energy by linking wave-energy physics, structurally efficient porous geometries, and robotic additive manufacturing. Ocean waves are a major untapped renewable energy resource capable of supplying up to 10% of global electricity (2,000 TWhr/year) while significantly reducing CO2 emissions. Yet offshore wave-energy approaches remain limited by high costs, difficult deployment, and fabrication constraints. To address these challenges, we develop meter-scale concrete modules that assemble into undulating seabed topographies derived from Bragg-resonance principles. This geometry focuses incoming waves toward a target region, increasing efficiency of downstream energy converters. This paper focuses on the fabrication-aware development of the modules. A unified design-to-production workflow translates the wave-lensing surface geometry into a porous, printable module by preserving the continuity of the sinusoidal top surface while reducing material volume. Six variants are generated through rotation, clipping, and scaling of gyroid Triply Periodic Minimal Surface (TPMS) lattices within a fixed envelope, then evaluated through metrics for solid top-surface percentage, number of top-surface openings, surface area, solid volume percentage, toolpath material volume, and extreme overhang percentage. Prototypes maintain approximately 80% of wave-facing surface continuity while reducing material volume by up to 60% compared to solid analogs. The modules are further assessed through toolpath and overhang analysis, tested as 1:10 scaled PLA fabrication models, and advanced through an initial concrete printing trial of the selected candidate. Results show that porosity cannot be evaluated through material reduction alone: the strongest candidate balanced high surface continuity with low 3-D printing overhang risk, while the most porous variant increased toolpath-material volume and fabrication risk. By translating a Bragg-resonance-derived seabed profile into a porous and printable module, this research advances a fabrication-ready pathway from wave-lensing geometry to experimentally testable offshore infrastructure, linking renewable energy generation to scalable, low-carbon marine construction.

Keywords: Structural morphology, architectural geometry, topology optimization, spatial structures, lattice structures, Triply Periodic Minimal Surfaces, digital fabrication, concrete printing, offshore structures.

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1. Introduction

Offshore wave-energy systems require large-scale structures that can operate in complex marine environments while remaining economical to fabricate, transport, and deploy. One strategy under investigation is the use of modular seabed topographies that alter wave focusing before energy conversion. Rather than concentrating only on the downstream energy converter, this approach shifts part of the design problem to the seabed surface itself: the geometry of the supporting infrastructure becomes an active component in shaping localized high-energy wave zones.

This research builds on prior work by Elandt, Shakeri, and Alam on surface-gravity-wave lensing [1], which demonstrates how periodic seabed topographies can focus wave energy through Bragg-resonance effects. In the present study, this work provides the target surface geometry: a sinusoidal seabed profile that is translated into the top-surface boundary of the repeatable modules.

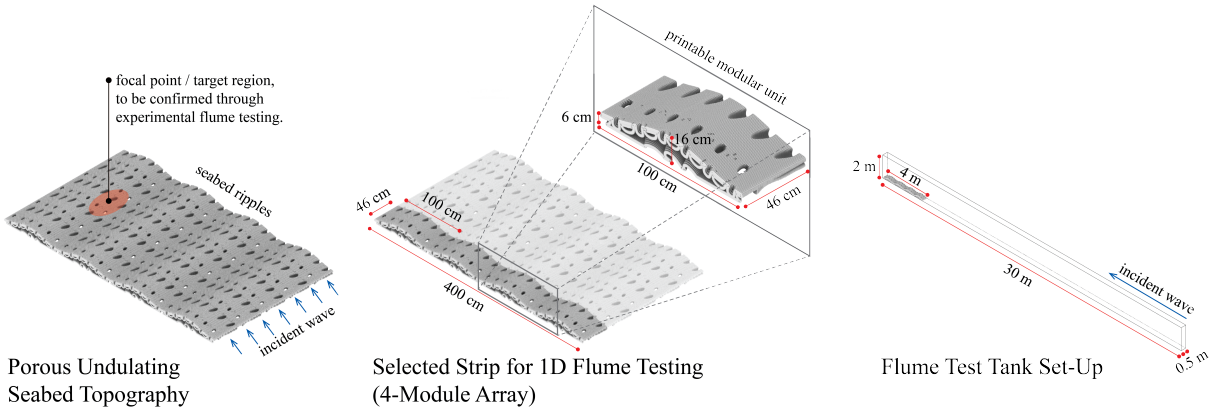


Figure 1: The broader project investigates High-Intensity Focused Ocean Waves (HIFOW) through modular, 3D-printed porous concrete seabed structures. The modules are intended to assemble into an undulating topography that generate resonant effects from incident waves and focus wave energy toward a target region.

The planned experiment consists of four 1-meter-long modules arranged as a continuous strip in a wave flume. Figure 1 situates this test within the broader HIFOW project, connecting the Bragg-resonance-derived seabed profile to the physical module array that will be fabricated and tested.

A solid concrete module would preserve the sinusoidal wave-facing surface most directly, but with high material volume, weight, and print time. To reduce these demands, the module interior is transformed using a gyroid Triply Periodic Minimal Surface (TPMS), a class of mathematically defined porous geometries characterized by continuous interconnected void networks. Unlike conventional infill strategies, gyroid TPMS structures provide tunable solid-void relationships while maintaining geometric continuity, making them especially suitable for additive manufacturing. Embedded within the sinusoidal envelope, the gyroid reduces solid volume while simultaneously altering the boundary conditions of the module.

The paper examines this coordination between the TPMS field and its sinusoidal envelope through six module variants. Their evaluation tests how porosity can be positioned so that the module remains legible as a wave-facing surface, efficient as a material system, and feasible as a printed object. The contribution of the paper is a fabrication-oriented method for translating a Bragg-resonance-derived seabed profile into porous TPMS-based modules, defining the conditions for the next phase of flume testing.

2. Research context

The design problem addressed in this paper emerges from the translation of a Bragg-resonance-driven wave-focusing seabed profile into a printable porous module. At the global scale, surface-gravity-wave

lensing defines the sinusoidal geometry of the seabed topography [1]. At the module scale, TPMS geometry introduces continuous porosity and material reduction. At the fabrication scale, robotic concrete printing imposes constraints on toolpath continuity, overhang, layer height, and material deposition. The module is therefore not only a wave-facing surface, a porous volume, or a printed object, but a negotiation between all three.

2.1. TPMS porosity as spatial and material strategy

Triply Periodic Minimal Surfaces (TPMS) provide the geometric basis for the continuous porous structures investigated here. Gyroid TPMS geometries are especially relevant for additive manufacturing because their continuous topology avoids isolated cavities, reducing discontinuities in material deposition. Previous research on gyroid cellular structures has shown how their properties can be adjusted through parameters such as volume fraction and material composition [2]. In architectural and structural design research, TPMS-based infills have also been developed as adaptive porous systems in which wavelength, thickness, orientation, and scalar-field modulation vary local density and surface condition [3]. In this paper, the gyroid is not treated as a generic infill within a neutral bounding volume. It is embedded in a sinusoidal envelope whose wave-facing surface must remain continuous enough for future testing. Its orientation, scale, and clipping therefore become design variables: they determine where openings occur, how much material is removed, and whether the geometry can be resolved through formwork-free layered deposition for fabrication.

2.2. Fabrication-aware robotic concrete printing

Robotic concrete printing introduces constraints that must be considered during geometric development. Layered extrusion requires attention to overhangs, path continuity, layer height, and estimated material volume. Recent work on automated toolpath design for 3D concrete printing frames toolpath generation as a design problem rather than a final slicing step [4]. Similarly, the surface-toolpath framework links surface and toolpath representations so that buildability and surface-quality constraints can feed back into geometric development [5]. The present work aligns with this fabrication-cognisant position. The toolpath is treated as an evaluative representation of the module, not only a production output.

3. Methodology

The methodology is organized as a controlled comparison: the global geometry is held constant across all variants, while the embedded gyroid field is varied. This allows differences in surface continuity, material volume, exposed area, and printability to be attributed to the TPMS-envelope relationship rather than to changes in the target seabed profile.

3.1. Module envelope and dimensional constraints

The module envelope is defined by the planned physical test strip. Each module preserves a 1 m wavelength derived from previous wave-lensing studies [1] and combines a 10 cm crest-to-trough sinusoidal relief with a 6 cm base, producing a total height of 16 cm. Four modules form the current 4m test strip. The envelope is held constant across all variants as a critical methodological constraint: the study does not compare different wave-facing profiles, but different ways of embedding porosity within the same profile.

3.2. Gyroid TPMS field variables

Within this fixed envelope, a gyroid TPMS field is embedded to introduce continuous porosity using the volumetric modeling tool Axolotl [6]. The field is varied through rotation, clipping, and scaling. Rotation changes the alignment between the TPMS field and the module boundaries; clipping defines which portions of the field remain within the envelope; and scaling in the X, Y, or global direction is used where necessary to maintain periodic continuity across adjacent modules.

3.3. Evaluation and fabrication-readiness workflow

The variants are evaluated through geometric and fabrication metrics defined in Section 4 and compared in Section 5. Selected variants are then translated into Ovenbird [4] toolpaths to evaluate layer organization, material-volume estimation, and overhang risk. 1:10 scaled fabrication models in PLA are used as initial physical demonstrations, and a concrete printing trial of the selected V3 candidate tests the transition from model-scale fabrication to concrete.

4. Evaluation metrics

The six variants are evaluated through metrics that correspond to the central tradeoff of the paper: how to introduce TPMS porosity without compromising the wave-facing surface or producing an unprintable geometry. The metrics are used comparatively to describe surface continuity, material reduction, geometric complexity, and fabrication risk.

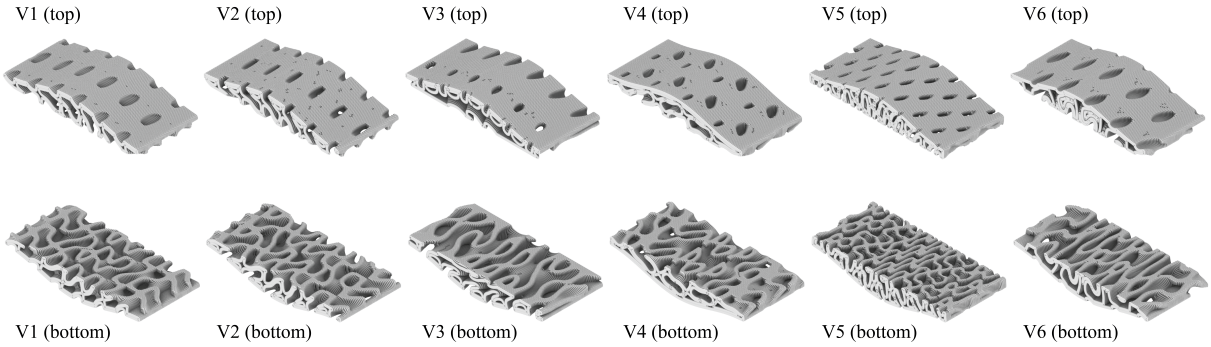


Figure 2: Six TPMS-based module variants generated within the fixed sinusoidal envelope.

4.1. Solid top-surface percentage, α_S

Solid top-surface percentage, α_S, measures the proportion of the wave-facing top surface that remains solid after the TPMS field is clipped by the module envelope:

$$\alpha_S = \frac{A_{\text{solid top}}}{A_{\text{total top}}}$$

The goal is to maximize α_S. A higher value indicates greater continuity of the target sinusoidal surface; a lower value indicates greater top-surface perforation. In this study, α_S is used as a proxy for preserving the wave-facing profile prior to hydrodynamic testing.

4.2. Number of surface holes, n

The number of surface holes, n, counts the openings created where the TPMS field intersects the top surface. It is treated as a descriptive metric rather than a direct optimization target. Used alongside α_S, it helps distinguish variants with similar solid top-surface percentages but different perforation patterns.

4.3. Solid volume percentage, α_V

Solid volume percentage, α_V, measures the proportion of the module's bounding volume occupied by solid geometry:

$$\alpha_V = \frac{V_{\text{solid}}}{V_{\text{bounding}}}$$

A lower value indicates greater nominal porosity and lower geometric density. However, α_V does not describe fabrication demand, so it is evaluated alongside toolpath-estimated material volume V. α_V is treated as a descriptive metric, while V expresses the actual material usage implications.

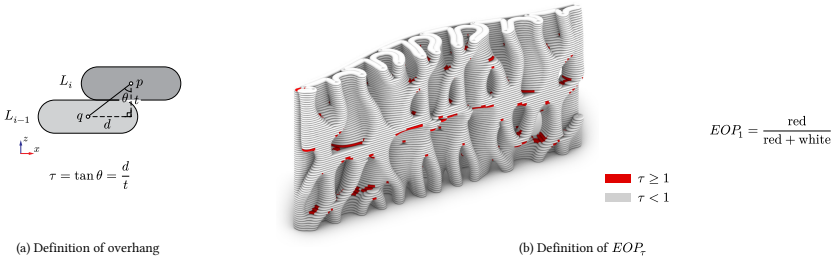
4.4. Material volume, V

Material volume, V, estimates the amount of material required to fabricate the module through robotic extrusion. Unlike α_V, which is calculated from the volumetric geometry, V is estimated from the toolpath length. The goal is to minimize V to reduce material use, weight, and print time. Comparing α_V and V helps distinguish nominal porosity from actual fabrication demand.

4.5. Total exposed surface area, S

Total exposed surface area, S, measures the area of all exposed surfaces in the module, including exterior boundaries and internal surfaces created by the porous network. It is treated as a descriptive metric rather than a direct optimization target. Higher values indicate greater internal articulation and surface exposure, which may be relevant for future fluid exchange or habitat-oriented studies, but may also correspond to increased geometric an fabrication complexity.

4.6. Extreme overhang percentage, EOP_1



(a) Definition of overhang

(b) Definition of EOP_τ

Figure 3: Definition of overhang and extreme overhang percentage.

Overhang τ measures how unsupported a sample point on a toolpath curve is (Figure 3a). Extreme overhang percentage, EOP_τ, records the percentage of the geometry exceeding an overhang limit of τ (Figure 3b). As the overhang of 1, or the overhang angle of 45^{deg} is seen as an upper limit of printable overhang, we use EOP_1 as a metric for minimization. A lower value indicates fewer overhang-sensitive regions and lower anticipated printability risk. In this study, EOP_1 is used comparatively; it does not guarantee full-scale concrete-printing success, which will also depend on material rheology and curing behavior.

5. Comparative analysis

Table 1: Quantitative evaluation of six variants. α_S–Solid top-surface percentage, n–Number of surface holes, α_V–Solid volume percentage, V–Material volume, S–Total surface area, EOP_1–Extreme overhang percentage.

Variant	α_S (m^2)	n	α_V	V (m^3)	S	EOP_1
	maximize	-	-	*minimize*	-	*minimize*
V1	79.25	12	40.74	0.037	2.03	0.063
V2	82.96	12	40.13	0.039	2.13	0.059
V3	84.22	12	40.56	0.041	2.17	0.027
V4	84.54	16	41.18	0.042	2.26	0.052
V5	80.63	28	37.09	0.049	2.72	0.071
V6	79.14	10	40.62	0.039	2.06	0.030

The six variants shown in Figure 2 were evaluated to test how changes in TPMS affect the module. The comparison is not intended to identify the most porous option, but to determine which configuration best negotiates the competing requirements of surface continuity, material reduction, and printability. The results are summarized in Table 1.

Table 2: Min-max normalized results. α_S–Solid top-surface percentage, V–Material volume, EOP_1–Extreme overhang percentage.

Variant	α_S (m^2)	-V (m^3)	-EOP_1
V1	0.020	1	0.182
V2	0.707	0.833	0.273
V3	0.941	0.667	1
V4	1	0.583	0.432
V5	0.276	0	0
V6	0	0.833	0.932

Because the metrics have different units and optimization directions, the performance metrics were normalized using min–max normalization to a range of [0,1]. For minimization objectives, the normalized values were inverted such that higher normalized values indicate improved performance. These scaled results are provided in Table 2 and visualized in Figure 4. Descriptive metrics are reported in Table 1 but excluded from Table 2 and Figure 4.

For maximization criteria:

$$x_i* = \frac{x_i - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}}$$

For minimization criteria:

$$x_i* = 1 - \frac{x_i - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}}$$

where x_i* is the normalized performance value.

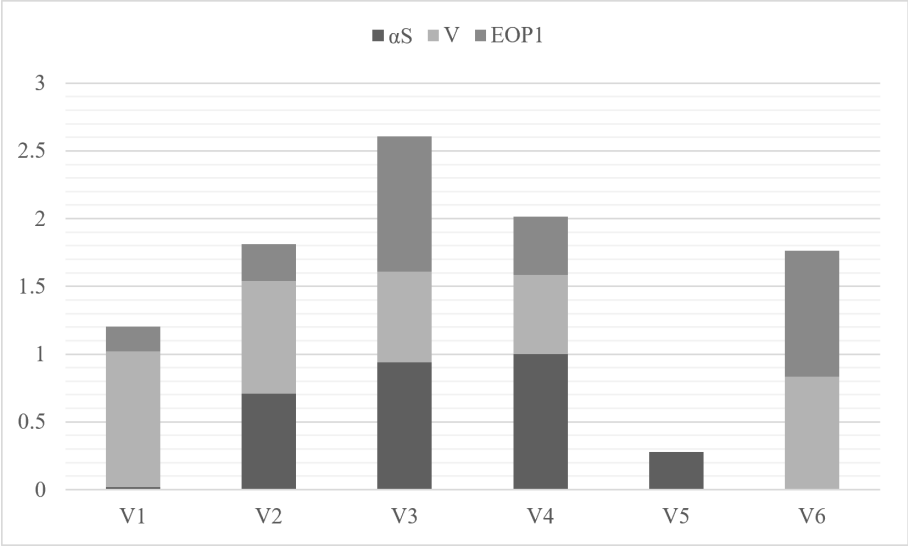


Figure 4: Normalized chart of six TPMS module variants.

5.1. Surface continuity and overhang risk

The percentage of solid top-surface, α_S, evaluates how much of the target sinusoidal wave-facing surface remains solid after the TPMS field intersects the envelope. V4 achieves the highest α_S at 84.54%, followed closely by V3 at 84.22%. If surface continuity were the only criterion, V4 would be the preferred option. However, the overhang metric changes this reading. V3 has the lowest EOP_1 value among all variants, 0.027, compared to 0.052 for V4. This distinction is significant because the same TPMS-envelope intersections that preserve or perforate the top surface also generate overhang-sensitive regions. Therefore, V3 provides the strongest balance between maintaining the target surface and reducing fabrication risk.

5.2. Material efficiency and fabrication demand

Volume-related metrics show that nominal porosity does not directly correspond to fabrication efficiency. V5 has the lowest solid volume percentage, α_V, at 37.09%, indicating the greatest volume reduction. However, it also produces the highest total exposed surface area, printing material volume, number of surface holes, and EOP_1 value. Therefore, V5 serves as a limit case: increasing porosity can also increase surface complexity, toolpath length, and overhang-sensitive geometry.

V1 presents a different tradeoff. It has the lowest toolpath-estimated material volume, at 0.037 m^3, and was initially preferred for its aligned layer transitions, clear opening logic, and clean layer organization. However, it also has one of the lowest solid top-surface percentages and a relatively high EOP_1 value. V1 is efficient in material-volume terms, but less competitive once surface continuity and overhang risk are factored in.

5.3. Candidate selection

V3 is selected as the current candidate for fabrication because it performs consistently across the criteria most critical to the next phase, as seen in Figure 4. It does not maximize a single isolated objective; rather, it preserves a high percentage of the wave-facing surface while minimizing overhang-related fabrication risk and reducing printing material quantity.

The six-variant study therefore does not produce a universal optimum. Instead, it clarifies the design space. V4 defines the upper range of top-surface continuity, V1 remains relevant as a low-material-volume and visually coherent option, and V5 demonstrates the fabrication penalties that can emerge when porosity is pushed too aggressively. The evaluation shows that porous TPMS modules must be judged through coupled criteria: where the voids occur, how the surface is preserved, and how the geometry resolves into a printable toolpath.

6. Toolpath generation and fabrication trials

In a TPMS-based module, the sinusoidal surface, module apertures, and boundary conditions are all affected by print orientation, layer height, extrusion width, and overhang behavior. The fabrication workflow therefore evaluates whether the target surface remains continuous, whether the porous network remains open, and whether the geometry can be resolved as a repeatable printed unit.

6.1. Toolpath orientation and slicing

The module was oriented so that the toolpaths run along the Y direction, parallel to the sinusoidal variation of the wave-facing surface. This orientation allows each printed stripe to follow the curvature of the target profile, rather than reconstructing the wave-facing surface through stepped layer increments. Print direction therefore becomes part of the surface-resolution strategy. While the paper does not test wave-focusing performance, it does evaluate whether the target surface can be maintained through fabrication. Printing along the sinusoidal direction reduces the risk that the wave-facing profile is geometrically degraded by layer segmentation.

6.2. Toolpath analysis and overhang behavior

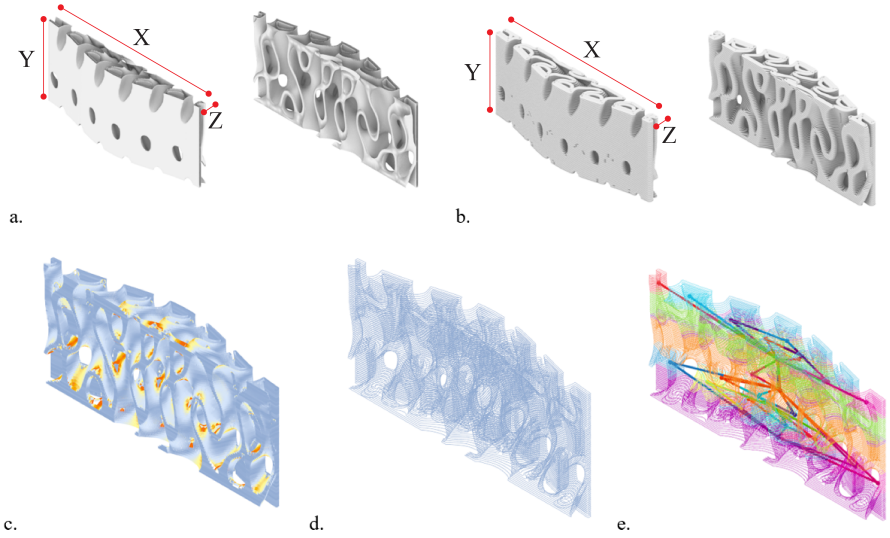


Figure 5: Analysis of V3 module using Ovenbird [4]: (a) Input mesh; (b) sliced mesh; (c) local overhang, heat map shows high-risk areas in red; (d) global overhang; and (e) toolpath topology.

The selected candidate was translated into Ovenbird toolpaths to evaluate the module as a sequence of deposited paths rather than as an ideal mesh. EOP_1 is used to compare the proportion of overhang-sensitive regions across variants. In this context, overhang analysis is not a pass/fail printability check; it identifies where the TPMS-envelope intersection produces fabrication risk. The same geometric operation that creates porosity also creates unsupported or difficult-to-resolve features. Toolpath analysis makes the fabrication consequences of TPMS orientation known before physical printing.

6.3. Scaled PLA fabrication models

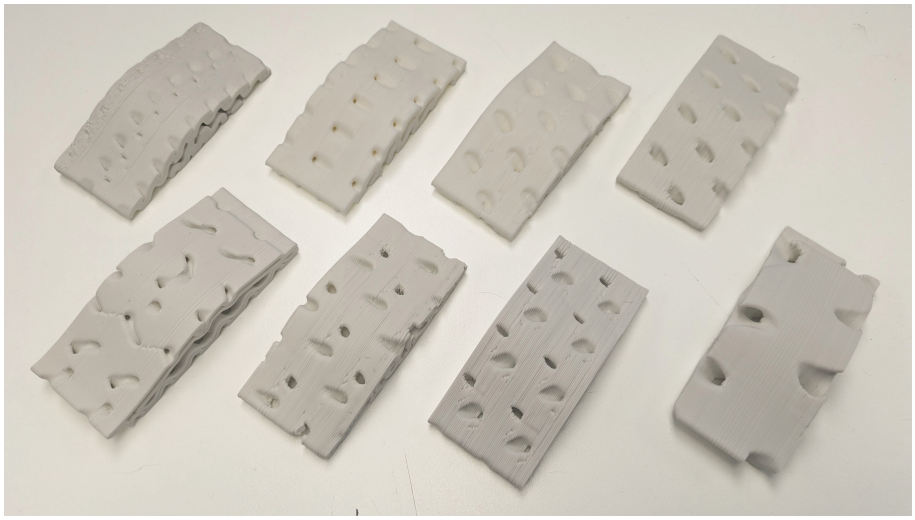


Figure 6: Scaled fabrication models.

Early design explorations were printed as scaled fabrication models [7] to verify the buildability and surface quality of the module design and toolpath scheme (Figure 6). The 1:10 scaled PLA models used a 0.7 mm layer height and a 2.1 mm print head, corresponding to the layer-based logic of the full-scale

concrete-printing workflow, which assumes a 7 mm layer height. These models make visible the effects of print resolution on surface articulation, apertures, and boundary legibility.

The PLA prints confirmed that the geometry remains legible and is able to be fabricated without formwork. The sinusoidal top profile remained distinguishable, the side and underside conditions remained visible, and the TPMS-derived void network remained open at model scale. However, the openings became smaller than in the original mesh. This aperture reduction is a significant fabrication finding: the designed voids survive printing, but their final size depends on extrusion width, material spread, and print scale.

6.4. Pilot concrete printing

Figure 7: Concrete printing results.

A pilot full-scale concrete printing experiment of the selected V3 design was performed. The setup involves an ABB® IRB6640 robotic arm, a MAI®MULTIMIX-3D pump, and a 3D Potter® Mixcell extruder. Two-component (2K) material was used (Sikacrete®-752 F and Sika® Sigunit® L-5601 AF accelerator). The design was printed without significant deformation, resulting in the desired porous shape (Fig. 7). However, one small branch failed to build up since the bottom loops needed more time to solidify before they could support the following loops. Such intricate features/small branches need to be simplified in future experiments, but will not affect the quantitative measures significantly.

7. Discussion and future work

Surface-gravity-wave lensing shows that wave energy concentration can be addressed through seabed morphology: patterned bottom topographies can redirect incident waves toward a target region through Bragg-resonance effects [1]. This paper addresses the architectural and fabrication problem that follows from that principle. A wave-lensing seabed profile must be converted from a continuous surface into repeatable modules that can be printed, assembled, and tested. The contribution of this paper is a fabrication-oriented method for developing that profile into a porous TPMS-based concrete module, establishing the geometric and fabrication criteria needed before hydrodynamic validation: surface continuity, controlled porosity, toolpath design, and module repeatability.

The results show that this conversion cannot be governed by material reduction alone. Within the fixed sinusoidal envelope, the gyroid TPMS field changes both the volume and the boundary of the module. The most porous variant did not produce the most fabrication-ready module; instead, it increased toolpath material volume and overhang risk. V3 is selected because it better balances surface continuity with printability. The evaluation therefore defines porosity as a coordinated geometric and fabrication condition, rather than an isolated measure of geometric volume reduction.

The concrete trial shifts the work from geometric selection to construction refinement: V3 must now be developed for multi-module assembly and flume testing, followed by evaluation of wave focusing, cyclic structural response, marine durability, and ecological performance. In doing so, the research advances a path from surface-gravity-wave lensing theory to scalable, low-carbon offshore infrastructure made from repeatable, 3D-printed porous concrete seabed components.

Acknowledgements

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